

Assignment: Strings in C Programming

1. Find the Length of a String Using strlen()

```
#include <stdio.h>

#include <string.h>


int main()
{
    char songTitle[] = "Tum Hi Ho";

    printf("Song Title: %s\n", songTitle);
    printf("Length of String: %lu\n", strlen(songTitle));

    return 0;
}
```

Sample Output

Song Title: Tum Hi Ho

Length of String: 9

2. Compare Two Usernames Using strcmp()

```
#include <stdio.h>

#include <string.h>


int main()
{
    char username1[50];
    char username2[50];
```

```
printf("Enter First Username: ");  
scanf("%s", username1);  
  
printf("Enter Second Username: ");  
scanf("%s", username2);  
  
if(strcmp(username1, username2) == 0)  
{  
    printf("Usernames are the same.\n");  
}  
else  
{  
    printf("Usernames are different.\n");  
}  
  
return 0;  
}
```

Sample Output

Enter First Username: virat18

Enter Second Username: virat18

Usernames are the same.

3. Copy "Flipkart" into shoppingApp Using strcpy()

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
{
    char shoppingApp[20];

    strcpy(shoppingApp, "Flipkart");

    printf("Shopping App: %s\n", shoppingApp);

    return 0;
}
```

Sample Output

Shopping App: Flipkart

4. Generate a Username from Full Name

```
#include <stdio.h>
#include <string.h>

int main()
{
    char fullName[50];
    char username[6];

    printf("Enter Full Name: ");
    fgets(fullName, sizeof(fullName), stdin);
```

```
fullName[strlen(fullName, "\n")] = '\0';

if(strlen(fullName) < 5)
{
    strcpy(username, fullName);
}
else
{
    strncpy(username, fullName, 5);
    username[5] = '\0';
}

printf("Generated Username: %s\n", username);

return 0;
}
```

Sample Output 1

Enter Full Name: Virat Kohli

Generated Username: Virat

Sample Output 2

Enter Full Name: Om

Generated Username: Om

Conclusion

This assignment demonstrates:

- Using strlen() to find string length.

- Using `strcmp()` to compare strings.
- Using `strcpy()` to copy strings.
- Using `strncpy()` to extract a portion of a string.
- Creating a simple username generator using string functions in C.